# Pabna University of Science & Technology

Department of Computer Science & Engineering
3rd Year 2nd Semester Examination-2025
Course Title **System Analysis and Design**
Course no: **CSE 3201**
Time: 3:00 hours (For PART-A and PART-B)

## PART-A

Full Marks: 35

N.B. (i) Answer any **Three** questions out of Four
(iii) Figures in the right margin indicate marks.

| | | | |
|---|---|---|---|
| 1 | (a) | What is System? Explain the characteristics of a system. | 5 |
| | (b) | Mention the elements of a system? Distinguish between: (i) Physical and Abstract System (ii) Open and Closed System | 4 |
| | (c) | What do you mean by Formal system and Informal system? | 2.67 |
| 2 | (a) | What is man-made information system? Classify in brief. | 3.67 |
| | (b) | Information needs change based on management levels. Describe the type of information required by Strategic (Top), Tactical (Middle) and Operational (Lower) management. | 4 |
| | (c) | Every system operates within an "Environment". Identify three external environmental factors that can impact an organization's information system design. | 4 |
| 3 | (a) | List the standard phases of the System Development Life Cycle (SDLC) in their correct chronological order. | 4.67 |
| | (b) | Mention the root causes for Project termination and Project failure. | 7 |
| 4 | (a) | What is feasibility study? What is the output of a feasibility study explain in brief? | 5.67 |
| | (b) | What are the works normally carried out at the phase Post implementation and maintenance phase? | 4 |
| | (c) | Explain Recognition of needs. | 2 |

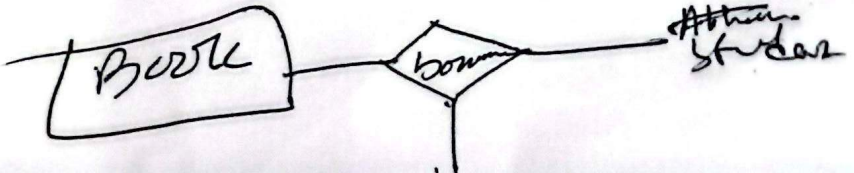
# Pabna University of Science & Technology

Department of Computer Science & Engineering
3rd Year 2nd Semester Examination-2025
Course Title **System Analysis and Design**
Course no: **CSE 3201**
Time: 3:00 hours (For PART-A and PART-B)

**PART-B** Full Marks: 35

N.B. (i) Answer any **Three** questions out of Four
(iii) Figures in the right margin indicate marks.

| | | | |
|---|---|---|---|
| 5 | (a) | What kinds or categories of information do you need when going to automate a manual system? Explain. | 3.67 |
| | (b) | Mention the primary internal and external sources from where the information originates. | 2 |
| | (c) | Explain different of Observations. | 6 |
| 6 | (a) | What are the primary advantages and disadvantages of Interview? Explain | 4.67 |
| | (b) | Differentiate between Interviews and Questionnaires. | 4 |
| | (c) | Explain different varieties of Closed questions. | 3 |
| 7 | (a) | What do you mean by structured analysis? What are the tools used for structured analysis? | 3.67 |
| | (b) | Mention the pros and Cons of different structured analysis tools. | 3 |
| | (c) | What are the skills a system analyst must have? Explain in brief | 5 |
| 8 | (a) | You are tasked with integrating the Inventory System of a university bookstore with the Library Management System. What shared data elements (e.g., ISBN) will act as the "Bridge"? If a student loses a library book, how can the system automatically check the Bookstore's Inventory to find the replacement cost and add it to the Student's Bill? | 5.67 |
| | (b) | You have designed a high-performance E-commerce system for a retail company. During the testing phase (UAT), the system worked perfectly with 50 simultaneous users. However, on "Black Friday," when traffic spiked to 5,000 simultaneous users, the system didn't just slow down-it completely crashed. The investigation reveals that while the CPU usage on the web server was only 30%, the Database Server was at 99% utilization, and the average response time for a product search increased from 200ms to 45 seconds before the connection timed out.<br>Based on the metrics provided (low CPU on web server, maxed-out DB), is this a Hardware problem (not enough RAM/CPU) or a Software/Design problem? Justify your answer using the concept of "Database Contention." | 6 |



# Pabna University of Science & Technology

Department of Computer Science & Engineering

3rd Year 2nd Semester Examination-2025

Course Title: Operating System

Course no: CSE 3203

Time: 3:00 hours (For PART-A and PART-B) || Full Marks:70

## PART-A

N.B. (i) Answer any **Three** questions from Part A
(iii) Figures in the right margin indicate marks.

Marks: 35

1 (a) Operating system is supposed to make efficient use of the computing hardware. However, sometimes it is appropriate for the operating system to forsake this principle and to "waste" resources. Is such system really wasteful or not? Justify your answer. 4

(b) How operating system protect itself from 3rd party malicious code execution? Is there any role of kernel mode-user mode provision in achieving such protection? Explain. 4

(c) Consider you are writing an operating system for a real-time fire detection system. Mention at least one difficulty that you must overcome. 3.67

2 (a) Imagine you purchased a new computer with Linux operating system pre-installed. When you turn on the computer for the first time, what operations it will perform during the boot process? 4

(b) Explain with a real-life scenario when you need to use virtual machine. Is there any alternative solution to provide similar benefits? 4

(c) "System calls are integral part of modern operating system" – to what extent do you agree or disagree with this statement? 3.67

3 (a) Is it possible to obtain a statistical profile of the amount of time a program spends executing different sections of its code? Briefly discuss the importance of obtaining such statistical profile. 4

(b) Suppose you are designing an operating system to be deployed in an IoT based home security system where multiple processes need to communicate. In this context, briefly discuss the strength and weakness of any two models of interprocess communication. 4

(c) What actions are taken by kernel to context switch between processes? 3.67

4 Consider the following set of processes, with the length of the CPU burst time given in milliseconds:

| Process | Burst Time | Priority |
|---|---|---|
| P1 | 2 | 2 |
| P2 | 1 | 1 |
| P3 | 8 | 4 |
| P4 | 4 | 2 |
| P5 | 5 | 3 |

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5 all at time 0.

(a) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a larger priority number implies a higher priority), and RR (quantum = 2). 4

(b) What is the turnaround time of each process for each of the scheduling algorithms in question no a)? 3

(c) What is the waiting time of each process for each of these scheduling algorithms? 2

(d) Which of the algorithms results in the minimum average waiting time (over all processes)? 2.67

## PART-B

Marks: 35

N.B. (i) Answer any **Three** questions from Part B
(iii) Figures in the right margin indicate marks.

5 (a) Do you think, studying different synchronization tools is beneficial for operating system engineering? How? 4

(b) Consider the following set of processes. 4

| Process | Brust Time | Arrival Time |
|---|---|---|
| P1 | 6 | 0 |
| P2 | 8 | 1 |
| P3 | 7 | 2 |
| P4 | 3 | 3 |

Apply Round Robin scheduling with quantum 4 and 8. Calculate the waiting time and propose which quantum time is better.

(c) To solve dining-philosophers, which is better - semaphore or monitor? 3.67

6 (a) Consider the given code and explain in which part of memory (stack, heap, data segment) will each of the variables be stored. 4

```
#include <stdio.h>
#include <stdlib.h>

int count=0;
int main()
{ int a=5, *ptr;
ptr = (int*) malloc (sizeof(int)); }
```

(b) Consider a system consisting of four resources of the same type (R1, R2, R3, R4) that are shared by the three processes (P1, P2, P3), each of which needs at most two resources. Show that the system is deadlock free. 3.67

(c) Explain the following terms- 4
i) Paging
ii) Fixed and Dynamic memory partitioning
iii) Fragmentation

7 (a) Given six memory partitions of 300 KB, 600 KB, 350 KB, 200 KB, 750 KB, and 125 KB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of size 115 KB, 500 KB, 358 KB, 200 KB, and 375 KB (in order)? 4

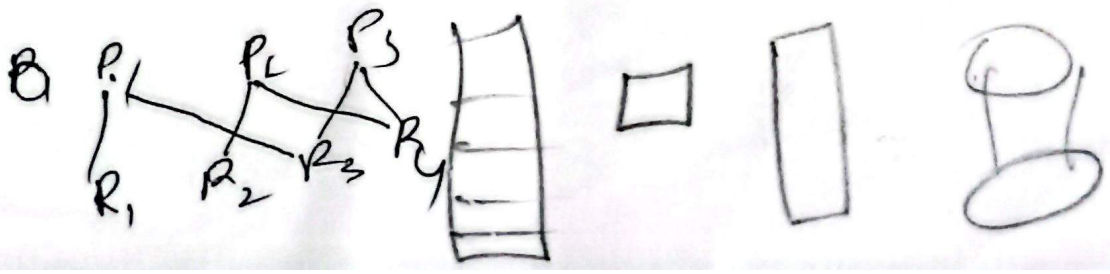
(b) Consider the following page reference string: 4
7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0, 1
Assuming demand paging with 3 frames, how many page faults would occur for the following replacement algorithms?
i) LRU replacement ii) FIFO replacement

(c) In which situation, virtual memory mechanism must be active in an operating system? 3.67

8 (a) "A file is a named collection of related information that is recorded on secondary storage" – to what extent do you agree or disagree with this statement? 4

(b) Consider you have 16 GB of RAM. You are trying to read a segment from a file of 100 GB. How conventional operating systems facilitate this operation? 4

(c) The list of all passwords usually kept in the operating system locally. In a multi-user system, how operating system provide safety to its user data so that one user cannot expose/exploit other users data? 3.67



# Pabna University of Science and Technology

## Department of Computer Science and Engineering

## B.Sc. Engineering 3rd Year 2nd Semester Examination- 2025

**Course Code:** CSE 3205 **Course Title:** Web Engineering

Time: 3 Hours Full Marks: 70

### PART-A

Full Marks: 35

N.B. (i) Answer any Three questions out of Four

(ii) Figures in the right margin indicate marks.

1. a) Define Web Engineering. Differentiate it from traditional software engineering. 2+2

   b) Categorize web applications based on development history and complexity. $2\frac{2}{3}$

   c) Describe four umbrella activities in Web Engineering. How would these activities ensure the success of a high-traffic site like the Bangladesh SSC Result Portal during result publication week? 2

   d) Summarize three Web Engineering best practices and justify how they contribute to the success of a large-scale project like Bkash (mobile financial services). 3

2. a) Draw a pictorial view (layered diagram) of the activities and actions involved in the Web Engineering process. 5

   b) Pabna University of Sciecne and Technology wants to develop a "Centralized Alumni Platform". The development team is in the Communication Phase. Stakeholders have conflicting requirements: $6\frac{2}{3}$

   - Alumni: Want Facebook/Google login for ease.
   - University Admin: Want national ID (NID) verification for security.
   - IT Department: Want open-source technology to avoid licensing costs.

   Describe how the team would conduct the formulation and elicitation actions to define the project's needs and scope.

3. a) Define Formulation. What are the Elicitation tasks. 1+2

   b) Enlist two internal and two external constraints for a web project developing a "Smart Grid Energy Trading Platform" for rooftop solar panel owners in Pabna (using blockchain for peer-to-peer energy trading). 2+2

   c) Biplob and Masud are managing a charity donation platform (online) development. They face conflict during Communication Phase: $4\frac{2}{3}$

   - Marketing team: One-click donation (no account) with 50% higher conversion data
   - Compliance team: Mandatory user registration for tax receipts due to 30% fake donations without registration
   - Development team: Supporting both workflows would cause a 5-week delay

   As project managers, propose two compromise solutions and one metric to measure negotiation success.

4. a) How The best negotiators strive for a win-win result? 5

   b) "PithaPedia" , a food delivery startup by Abu Sayed focusing on traditional Bangladeshi pithas, is developing a mobile app to connect home chefs with customers. The system must handle: $6\frac{2}{3}$

   i. Order Placement: Customers select pithas (e.g., Bhapa, Chitoi), apply coupons, and confirm payment via mobile banking.

   ii. Order Processing: Home chefs receive orders, update status (e.g., "Steaming," "Out for Delivery").

   iii. Real-Time Tracking: Customers and delivery riders track order location via GPS.

   Create a sequence diagram for the "Order Placement" workflow to clarify system interactions.

## PART-B

N.B. (i) Answer any Three questions out of Four
(ii) Figures in the right margin indicate marks.

Full Marks: 35

5. a) What are the outputs of analysis phase? 1+2
   b) What are the common questions that should be answered to find user class? 2
   c) A startup is developing a "Poultry Farm Management" WebApp for small-scale farmers in Mymensingh. The system must cater to three primary user roles: $6\frac{2}{3}$
      i. Farmers (track flock health, feed inventory, egg production, get disease alerts)
      ii. Veterinarians (view farm logs, prescribe medicines remotely, schedule visits)
      iii. Feed Suppliers (manage stock, receive bulk orders, track deliveries)

      List all primary/secondary actors and at least 4 core use cases (Example: "Log Egg Production") for this WebApp.

6. a) Enlist the four design goals for an AI-powered voice assistant app targeting illiterate rural mothers. 4
   b) Draw the hierarchy of design actions. 3
   c) A group of students want to develop a "PWA for Pabna Artisans" (weavers in Pabna and Sirajgoinj). The system must serve three user roles: $4\frac{2}{3}$
      i. Artisans (offline product catalog upload, offline order management, sync when internet available, receive SMS when new orders arrive)
      ii. Buyers (Urban/Dhaka) (browse products online, place orders, track shipment – no offline needed)
      iii. Fair Trade Cooperative Admins (approve artisan registrations offline (sync later), verify product authenticity, release payments via bKash/Nagad)

      Create a activity diagram for the "Order Placement" workflow to clarify system interactions.

7. a) Enlist the generic questions, that should be answered for a well-designed interface. 3
   b) Justify the statement: *"Adapt to the problem, to the project, to the team, and to the organizational culture."* 3
   c) A group of students want to develop a "Campus Event Manager" WebApp for Pabna University of Science and Technology. The system must serve three user roles: $5\frac{2}{3}$
      - Students: Browse/register for events (Mukto Moncho, Workshop), view schedules, receive reminders.
      - Event Organizers (Clubs): Create events, approve registrations, track attendance.
      - Administrators (Proctor Office): Manage user permissions and generate reports.

      Design a low-fidelity mockup for the student-facing interface.

8. a) How do you recognize a "Good" WebE team? 3
   b) Write down the appropriate way to assess quality of web application. 3
   c) Online blood management system is a web application that is usually designed for the people who are needed to donate blood or those who need to get blood. All the information will provide in this application like blood group, donor information, and hospital/location information that they will easily connect to each other. If any person wants to donate the blood so he needs to get the registration then he will be able to donate the blood and became the member of this application. It provides the following functions Admin, Donor, Donor Registration, Acceptors, Donor Search, and Acceptors Review. $5\frac{2}{3}$

      Now, you have to design logical and physical design paradigm for the system.

# Pabna University of Science and Technology
## Department of Computer Science and Engineering
## B. Sc. Engg. 3rd Year 2nd Semester Final Examination 2024

**Course Code:** CSE-3207 **Course Title:** Digital Signal Processing
**Time:** 3 Hours (Part-A and Part-B) **Session:** 2021- 22 **Full Marks:** 35+35=70
[Figures in the right margin indicate full marks. Answer **Six** questions taking **Three** from each part]

### PART-A

1. (a) Define digital signal. What are the advantages and limitations of digital signal processing? 4
   (b) Differentiate among analog signal, digital signal and discrete signal. 4
   (c) Why should we study DSP? What are the applications of it? $3\frac{2}{3}$

2. (a) Sketch the following signals: 4
   $(i) x(t) = \prod(2t+3)$ (ii) $x(t) = 2\prod(t-\frac{1}{4})$
   (iii) $x(t) = \cos(20\pi t - 5\pi)$ (iv) $x(t) = r(-0.5t+2)$
   Here, $\prod()$ defines unit pulse and r() defines ramp signal.
   (b) Define sampling and sampling rate. Also explain quantization error and aliasing. $3\frac{2}{3}$
   (c) What do you understand by cross-correlation of two signals? What is the physical significance of cross-correlation? 4

3. (a) Define z transform, inverse z transform, and ROC of z transform. 3
   (b) Determine the z transform and ROC of the following signals: 4
   (i) $x(n) = \{2, 4, 5, 7, 0, 1, 3\}$ (↑ at 0)
   (ii) $x(n) = \delta(n+1) + 3\delta(n) + 6\delta(n-3) - \delta(n-4)$
   (c) By using partial fraction expansion method, find the inverse z transform of $4\frac{2}{3}$

$$H(z) = \frac{-4 + 8z^{-1}}{1 + 6z^{-1} + 8z^{-2}}$$

4. (a) Determine if the systems described by the following are linear or non-linear $3\frac{2}{3}$
   (i) $y(n) = Ax(n) + B$
   (ii) $y(n) = x^2(n)$
   (b) Compute the convolution $x(n)$ of the signals 4

$$x_1(n) = \{1, -2, 1\}$$
$$x_2(n) = \begin{cases} 1, & 0 \le n \le 5 \\ 0, & elsewhere \end{cases}$$

   (c) A biomedical monitoring system acquires an analog ECG signal with a maximum frequency component of 150 Hz. The signal is converted into digital form for computer-based signal analysis and storage. 4
   Now, determine the minimum sampling frequency required to digitize the ECG signal without aliasing. Justify your answer mathematically.

**Pabna University of Science and Technology**
**Department of Computer Science and Engineering**
**B. Sc. Engg. 3rd Year 2nd Semester Final Examination 2024**

**Course Code:** CSE-3207 **Course Title:** Digital Signal Processing
**Time:** 3 Hours (Part-A and Part-B) **Session:** 2021- 22 **Full Marks:** 35+35=70
[Figures in the right margin indicate full marks. Answer **Six** questions taking **Three** from each part]

## PART-B

5. (a) Determine whether the following signals are energy signals, power signals, or neither. $3\frac{2}{3}$
   (i) $x(t) = e^{-at}\, u(t)$
   (ii) $x(t) = tu(t)$.
   (b) Draw the 8 point DIFFFT butterfly and obtain DFT of the given sequence, 5
   $$x(n) = \{1, 0, 1, 0\}$$
   (c) Differentiate between DFT and DTFT. Also discuss the advantages of FFT over DFT. 3

6. (a) What do you mean by Fourier Transform? Explain Gibbs Phenomenon. 3
   (b) A smartphone company is developing a noise-removal feature that separates a user's voice from repetitive background machine sound. The system first performs Fourier analysis and then separates components based on dominant Fourier coefficients. The following signal is applied to the system: $4\frac{2}{3}$

$$x(n) = \begin{cases} 2 & n \bmod 4 = 0 \\ 0 & n \bmod 4 = 1 \\ -2 & n \bmod 4 = 2 \\ 0 & n \bmod 4 = 3 \end{cases}$$

   Compute the Fourier series coefficients for one period and the dominant frequency component(s) suitable for separation.
   (c) Determine and sketch the energy density spectrum $S_{xx}(\omega)$ of the signal $x(n) = a^n u(n)$, $-1 < a < 1$. 4

7. (a) Prove the following properties: 4
   (i) Time reversal
   (ii) Parseval's theorem
   (b) Define Memory System and System Causality with proper example. 4
   (c) What is the need for short time spectral analysis? $3\frac{2}{3}$

8. (a) What is Odd and Even Signal? Find the odd and even components of the signal $x(t) = e^{-2t}\cos t$. 4
   (b) Write a MATLAB or python program that will find the convolution between two discrete time signals and also will plot them in suitable graphs. $3\frac{2}{3}$
   (c) What is a vocoder? Explain with a block diagram. 4

# Pabna University of Science and Technology

**Department of Computer Science and Engineering**

B. Sc. Engineering 3rd Year 2nd Semester Examination 2024

Course Title: Microprocessor and Assembly Language

Course Code: CSE 3209

Time: 3:00 hours (PART-A & PART -B) Full Marks: 70

**Marks 35**

**PART-A**

**N.B: (i) Answer any three questions from PART A (Q1 to Q4).**
**(ii) Figures in the right margin indicate marks.**

1 (a) State any three features of 8086 microprocessor that makes it differentiable from 8088 microprocessor. 3

(b) Write down the process of loading operating system when PC is powered up. Draw the block diagram of 8086 architecture. 6

(c) Suppose 8086 microprocessor has 16 bits address bus. How many memory bytes can be accessed? $2\frac{2}{3}$

2 (a) Explain **segmented memory architecture of 8086 and** critically analyze the advantages of segmentation and limitations compared to modern flat memory models. Then solve:
DS = 1000H, Offset = 0029H → Find physical address 4

(b) Explain pipelining in: 4
- 8086 (prefetch queue)
- RISC (5-stage pipeline)
- Pentium 4 (20–31 stage pipeline)

Then Identify **data, structural, and control hazards.**

(c) The **80386** introduced 32-bit architecture. Explain how this impacts data processing speed and memory addressing compared to 80286. $3\frac{2}{3}$

3 (a) A system requires running multiple programs simultaneously. Which processor (8086 / 80286 / 80386) would you choose and why? $2\frac{2}{3}$

(b) In an 8086 system: CS = 34BAH and IP = 8AB4H 4
Calculate the **physical address** of the next instruction and explain the following terms
I. Segment: Offset addressing
II. Role of address summing block

(c) In the context of 80386, explain how **segmentation and paging** work together for memory management. 3

(d) Suppose you need to design a system with **more than 1MB memory**. Explain why 8086 is insufficient and how 80286 solves this problem. 2

4 (a) Discuss the advantages and limitations of **Real Mode vs Protected Mode** in 80286. $2\frac{2}{3}$

(b) Compare **RISC vs CISC architectures** in terms of: Instruction execution time, Memory access pattern, Pipeline efficiency and Hardware complexity 5
Then **justify why modern processors (like Intel Pentium 4)** still use CISC but internally behave like RISC.

(c) Design a **hypothetical processor** by choosing: RISC or CISC, Number of registers, and Pipeline stages. 4
Now, justify your design choices based on: Performance, Power consumption and Complexity

## PART-B

**N.B: (i) Answer any three questions from PART B (Q5 to Q8).** **Marks 35**
**(ii) Figures in the right margin indicate marks.**

5. (a) A digital clock uses a seven-segment display. Explain how numbers are displayed using binary inputs. $2\frac{2}{3}$

   (b) Explain why **dynamic partitioning suffers from external fragmentation**, while paging does not. Evaluate the role of **virtual memory** in 80286. 4

   (c) Differentiate between **Carry Flag (CF)** and **Overflow Flag (OF)** using binary examples of unsigned and signed addition overflow 2

   (d) Write an assembly routine to determine whether a given byte value is even or odd without using DIV instruction. Explain why your logic works. 3

6. (a) Write an assembly program equivalent to the following C code and explain short-circuit logic implementation. $2\frac{2}{3}$

```
if(A>B && B>C)
    X=1;
else
    X=0;
```

   (b) A signed arithmetic routine adds two 8-bit numbers stored in memory. Design an 8086 program that detects: overflow, zero result, and Carry generation. Use appropriate flags and explain why each flag is checked. 3

   (c) *Explain the difference between signed and unsigned jumps using examples of JA, JG, JB, JL.* 3

   (d) *Analyze the logic error for the following assembly code, explain the final output for all cases, and rewrite the correct version.* 3

```
CMP AX, BX
JG L1
MOV CX, BX
L1: MOV CX, AX
```

7. (a) A program crashes due to stack overflow. Explain probable causes in assembly language and methods to prevent it. $2\frac{2}{3}$

   (b) Write assembly code to reverse bit pattern of an 8-bit number. 3

   (c) Differentiate arithmetic right shift and logical right shift using negative binary numbers. 3

   (d) Explain with example how XOR can be used for temporary encryption/decryption of stored data. 3

8. (a) Discuss different types of addressing mode with example. $2\frac{2}{3}$

   (b) Design a payroll system in assembly where employee basic salary array is processed and tax calculated using procedures, arrays, jumps, MUL/DIV. 5

   (c) Develop a string comparison routine using CMPSB that checks whether two passwords are identical. If mismatch occurs, display error. 2

   (d) Explain why DX must be cleared before unsigned DIV of 16-bit dividend. Find quotient, remainder, and affected registers for the following code. 2

```
MOV AX, 1234H
MOV BX, 0010H
DIV BX
```

# PABNA UNIVERSITY OF SCIENCE AND TECHNOLOGY
## Department of Computer Science and Engineering
## Pre-Lab Test-2023

Course Code: CSE-3210, Course Title: Microprocessor and Assembly Language Sessional

---

P-1: Create an assembly code to perform the addition of two single-digit inputs and display the result.
P-2: Write an assembly code that compares three values and identifies the highest.
P-3: Write an assembly program to multiply two numbers and present the result as a two-digit number.
P-4: Create an assembly code that prints the numbers 0 through 9 using only conditional jump instructions.
P-5: Develop an assembly language pattern program to generate a left-justified inverted triangle using *.
P-6: Develop an assembly language code to generate a triangle of asterisks aligned to the right.
P-7: Construct an assembly program to print all uppercase letters with a space before each letter.
P-8: Create an assembly language script that outputs the multiplication table for a specified number.
P-9: Write an assembly code to reverse a number using the stack with PUSH and POP instructions.
P-10: Create an assembly program to check if a given number or word is a palindrome and display the result.

# Microprocessor and Assembly Language Sessional

Total Marks 40 (5*8) || Time 2 hours

You have to write assembly code and simulate outputs step-by-step (register trace)

## Q1. Maximum Number from Array

Write an 8086 assembly program to:
- Store 5 unsigned 8-bit numbers in memory
- Find the maximum number and store it in a register or memory

Requirements:
- Use register indirect addressing
- Use LOOP instruction

Sample Input:
Array: 12H, 45H, 23H, 67H, 34H

Sample Output:
Maximum = 67H

## Q2. String Reverse using String Instructions

Write an 8086 program to:
- Reverse a string stored in memory

Requirements:
- Use SI and DI
- Use LODSB and STOSB

Sample Input:
String: HELLO

Sample Output:
Reversed String: OLLEH

## Q3. Address Calculation & Tracing

Analyze the following program:

```
MOV AX, 2000H
MOV DS, AX
MOV BX, 0100H
MOV SI, 0020H
MOV AX, [BX+SI]
ADD AX, 0010H
```

Tasks:
- Determine effective address (EA)
- Determine physical address
- Determine final value of AX

Given:
Memory[2000:0120] = 1234H

Expected Output:
EA = 0120H
Physical Address = 20120H
Final AX = 1244H

## Q4. Loop Execution Trace

Analyze the following program:

```
MOV CX, 0005H
MOV AX, 0000H

L1: ADD AX, CX
LOOP L1
```

Task:
- Find final value of AX

Expected Output:
AX = 000FH

# Microprocessor and Assembly Language Sessional

Total Marks 40 (5*8) || Time 2 hours

You have to write assembly code and simulate outputs step-by-step (register trace)

## Q5. Sum of Even Numbers in Array

Write an 8086 program to:
- Find the sum of even numbers from an array of 6 elements

Requirements:
- Use conditional jump
- Use loop

Sample Input:
Array: 02H, 05H, 08H, 03H, 06H, 01H

Sample Output:
*Sum = 10H*

## Q6. Factorial using Procedure

Write an 8086 program to:
- Calculate factorial using CALL and RET

Requirements:
- Use stack (PUSH/POP)

Sample Input:
N = 4

Sample Output:
Factorial = 24 (18H)

## Q7. Count Number of 1s in a Byte

Write an 8086 program to:
- Count number of 1 bits in a byte

Requirements:
- Use shift or rotate instructions

Sample Input:
Number = 5AH

Sample Output:
Number of 1s = 4

## Q8. Arithmetic Expression + 7-Segment Display

Write an 8086 program to:
- Compute Result = ((A × B) + C) / D
- Display result on 7-segment display

Requirements:
- Use MUL and DIV

Sample Input:
A = 02H, B = 03H, C = 04H, D = 02H

Sample Output:
Result = 05H

# Class Test-01

Course Code: CSE 3205 Course Title: Web Engineering Date: 06/12/2025 (SAT 09:30 AM)
3rd Year, 2nd Semester |Topics: Communication | Duration: 30 Minutes | Full Marks: 10

a) Define Communication. What are the Elicitation tasks. 1+1

b) Enlist the internal and external constraints in Web Project Development. 2

c) A university wants to develop a e-learning platform to facilitate online courses, student collaboration, and real-time assessments. The development team is currently in the Communication Phase of the Web Engineering Process. Stakeholders include professors, students, and IT administrators, each with conflicting requirements: 3

- Professors want advanced analytics to track student progress (desktop/laptop UI).
- Students demand an intuitive interface with mobile access.
- IT administrators prioritize security and scalability.

Describe how the team would conduct the **formulation** and **elicitation** actions to define the project's needs and scope.

d) How the best negotiators strive for a win-win result? 2

# Pabna University of Science and Technology
## Department of Computer Science and Engineering
## Class Test-1

Course Title: Digital Signal Processing — Time: 25 minutes
Course Code: CSE-3207 — Full Marks: 10

a. Define Signal and Systems with examples. 2

b. Determine whether the following signals are energy signals, power signals, or neither. 4

i) $X(t) = e^{-at}\, u(t)$

ii) $X(t) = tu(t)$.

c. Explain causal system. Determine if the system described by the following equation are linear or non-linear, $y(n) = Ax(n) + B$. 4

VD. Superposition.

**Set B: The Employee Commission Tracker**

**Problem Statement:**

A. Write a shell script to calculate the monthly commission for a sales executive.

1. Take the **Employee ID** and **Total Sales Amount** as input from the user. (2 Marks)
2. Calculate the commission based on the following tiers: (5 Marks)
    - Sales <= 5000: $0\%$ commission.
    - Sales > 5000 and <= 15000: $10\%$ commission.
    - Sales > 15000: $15\%$ commission.
3. If the commission is greater than $0$, deduct a **Processing Fee of 2.5%** from the commission amount. Use bc for precise calculation. (3 Marks)

B. write a program in any language to demonstrate the usage of fork system call. **10 Marks**

**Grading Rubric for A (Total: 10 Marks):**

- **2 Marks:** Correct use of read to capture ID and sales amount.
- **5 Marks:** Correct if-elif-else structure for commission tiers.
- **2 Marks:** Correct implementation of bc for decimal math (2.5% deduction).
- **1 Mark:** Output formatting (displaying the Net Commission).

**Set B: The Employee Commission Tracker**

**Problem Statement:**

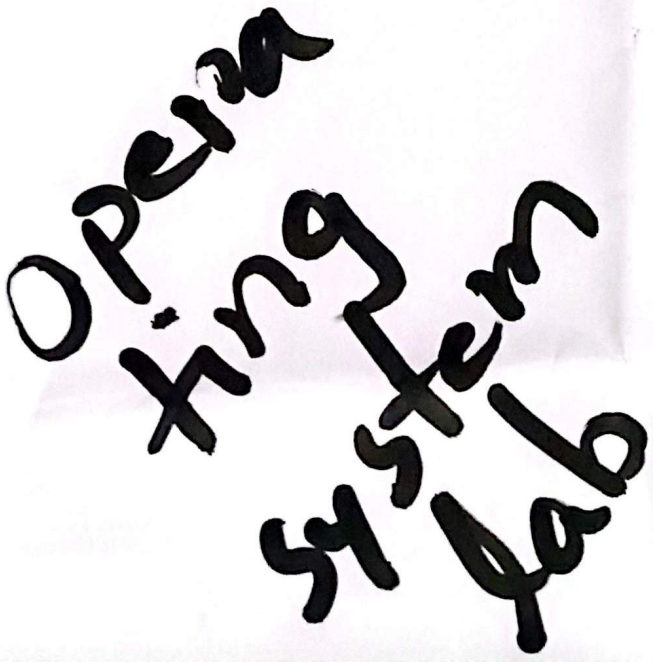
A. Write a shell script to calculate the monthly commission for a sales executive.

1. Take the **Employee ID** and **Total Sales Amount** as input from the user. (2 Marks)
2. Calculate the commission based on the following tiers: (5 Marks)
    - Sales <= 5000: $0\%$ commission.
    - Sales > 5000 and <= 15000: $10\%$ commission.
    - Sales > 15000: $15\%$ commission.
3. If the commission is greater than $0$, deduct a **Processing Fee of 2.5%** from the commission amount. Use bc for precise calculation. (3 Marks)

B. write a program in any language to demonstrate the usage of fork system call. **10 Marks**

**Grading Rubric for A (Total: 10 Marks):**

- **2 Marks:** Correct use of read to capture ID and sales amount.
- **5 Marks:** Correct if-elif-else structure for commission tiers.
- **2 Marks:** Correct implementation of bc for decimal math (2.5% deduction).
- **1 Mark:** Output formatting (displaying the Net Commission).



**Pabna University of Science and Technology**
**Department of Computer Science and Engineering**
**Class Test-2**

**Course Title: Digital Signal Processing** **Time: 25 minutes**
**Course Code: CSE-3207** **Full Marks: 10**

(a) Define Z-transform and Inverse Z-transform. 2
(b) State and prove time reversal and convolution property in z-domain. 4
(c) `Determine the inverse $z$-transform of $X(z) = \frac{1}{1-1.5z^{-1}+0.5z^{-2}}$ 4

if **(a)** ROC: $|z| > 1$
**(b)** ROC: $|z| < 0.5$
**(c)** ROC: $0.5 < |z| < 1$

$x(n) = \left(\frac{1}{2}\right)^n u(n)$

$x(n) = \delta(n+k): k>0$

OR

(a) What do you mean by Fourier Transform and Gibbs Phenomenon? 2
(b) State and prove correlation theorem and Parseval's theorem in frequency domain. 4
(c) Determine and sketch the energy density spectrum $S_{xx}(\omega)$ of the signal $x(n) = a^n u(n)$, $-1 < a < 1$. 4

# Final Lab Exam

Course Code: CSE 3206 Course Title: Web Engineering Sessional Date: 11/04/2026 (SAT 10:30 AM)
3rd Year 2nd Semester

## Lab Question – 1

Develop a Single Page Application (SPA) using the MERN stack (MongoDB, Express.js, React.js, Node.js) and AJAX that displays and manages a list of Top Bangladeshi EdTech Startups, similar to the functionality of a leaderboard or admin dashboard.

**General User Functionality:**

1. View a list of top Bangladeshi EdTech startups (name, sector, founder(s), headquarters, year founded, total funding, logo URL).
2. Implement a search box to filter the list by startup name or sector (e.g., K-12, test prep, upskilling, coding).
3. Data should be loaded dynamically via AJAX (no page reloads).

**Admin Panel Functionality:**

1. Admin must log in using a login form to manage the list.
2. Once logged in, admin should see:
   - *Add New Startup* button (opens a modal/form).
   - *Edit* and *Delete* buttons for each listed startup.
3. All actions (add/edit/delete) must be handled via AJAX requests.

**Technical Requirements:**

- *Frontend*: React.js (SPA with routing if needed)
- *Backend*: Express.js (REST API)
- *Database*: MongoDB (Use Mongoose for schema design)
- *Authentication*: JWT-based simple login system for admin
- *AJAX*: Use `fetch()` or Axios for all client-server interactions
- *Styling*: Tailwind CSS or Bootstrap (optional but encouraged)

**Admin Credentials (Hardcoded):**

- *Username*: `admin_edtech`
- *Password*: `admin@123`

**(Optional):**

- Implement pagination for large lists (e.g., 5 startups per page).
- Sort startups by total funding (highest to lowest) by default.
- Show a confirmation dialog before deleting a startup.

## Lab Question – 2 (Answer should be written in script)

- **Following are the requirements of a client (https://www.peopleperhour.com/freelance-jobs/technology-programming/website-development/research-project-groundwater-dashboard-4486849).**

- **Read the requirements carefully and prepare a short project proposal containing Timeline, Budget, Milestones and Deliverables.**

**Project overview**

The objective of this project is to develop a web application called Groundwater Dashboard Explorer (GWDE). The GWDE is a front-end web application designed to visualize geospatial ground water data (risks, opportunities, and climate scenarios) for a research project. The codebase must be reusable as a "blueprint" that can easily be forked and customized for other projects.

**Core Functionalities**

- **Geospatial Visualisation**
  Interactive maps (raster/shapefile) with pan/zoom, layer toggling, and zoom-dependent detail.

**Timeline & Deliverables (2026)**

May: Initial wireframes and user journey mock-ups.
August: Functional Beta version with test API integration.
October: Final launch, bug fixing, and codebase handover for deployment.

**Budget:** 20.000 Euro (non-negotiable)

**Technical Stack & Requirements**

- **Framework:** Vue.js with the Vuetify UI component library
- **Language:** TypeScript (required for clean, type-safe, and scalable code)
- **Architecture:** Clean Architecture principles (ensures the codebase can be easily forked and customised by IIASA researchers)
- **Rendering:** Familiarity with Server-Side Rendering (SSR) via Nuxt.js (required for performance and SEO optimisation)

**Geospatial & Data Visualization**

- **Mapping Libraries:** OpenLayers or Leaflet (must support Raster and Shapefile overlays)
- **Data Formats:** Integration with GeoTIFF (COG), NetCDF, GeoJSON, CSV, and potentially Zarr
- **Charting:** Interactive data plots using D3.js, Chart.js, Highcharts, or similar
- **Base Maps:** Support for background map toggling and zoom-dependent data rendering

**Infrastructure & Integration**

- **API Consumption:** Interfacing with IIASA's cloud-hosted REST API that will provide the data to be displayed
- **Styling:** Modern CSS (e.g., Tailwind CSS) for responsive, mobile-friendly design
- **Version Control:** Git-based workflow (hosted on IIASA repositories)

# Final Lab Exam

Course Code: CSE 3206 Course Title: Web Engineering Sessional Date: 11/04/2026 (SAT 10:30 AM)
3rd Year 2nd Semester

## Lab Question – 1

Develop a Single Page Application (SPA) using the MERN stack (MongoDB, Express.js, React.js, Node.js) and AJAX that displays and manages a list of Top Bangladeshi EdTech Startups, similar to the functionality of a leaderboard or admin dashboard.

**General User Functionality:**
1. View a list of top Bangladeshi EdTech startups (name, sector, founder(s), headquarters, year founded, total funding, logo URL).
2. Implement a search box to filter the list by startup name or sector (e.g., K-12, test prep, upskilling, coding).
3. Data should be loaded dynamically via AJAX (no page reloads).

**Admin Panel Functionality:**
1. Admin must log in using a login form to manage the list.
2. Once logged in, admin should see:
   - *Add New Startup* button (opens a modal/form).
   - *Edit* and *Delete* buttons for each listed startup.
3. All actions (add/edit/delete) must be handled via AJAX requests.

**Technical Requirements:**
- *Frontend*: React.js (SPA with routing if needed)
- *Backend*: Express.js (REST API)
- *Database*: MongoDB (Use Mongoose for schema design)
- *Authentication*: JWT-based simple login system for admin
- *AJAX*: Use `fetch()` or Axios for all client-server interactions
- *Styling*: Tailwind CSS or Bootstrap (optional but encouraged)

**Admin Credentials (Hardcoded):**
- *Username*: `admin_edtech`
- *Password*: `admin@123`

**(Optional):**
- Implement pagination for large lists (e.g., 5 startups per page).
- Sort startups by total funding (highest to lowest) by default.
- Show a confirmation dialog before deleting a startup.

## Lab Question – 2 (Answer should be written in script)

- **Following are the requirements of a client (https://www.peopleperhour.com/freelance-jobs/technology-programming/website-development/research-project-groundwater-dashboard-4486849).**
- **Read the requirements carefully and prepare a short project proposal containing Timeline, Budget, Milestones and Deliverables.**

**Project overview**

The objective of this project is to develop a web application called Groundwater Dashboard Explorer (GWDE). The GWDE is a front-end web application designed to visualize geospatial ground water data (risks, opportunities, and climate scenarios) for a research project. The codebase must be reusable as a "blueprint" that can easily be forked and customized for other projects.

**Core Functionalities**

**Geospatial Visualisation**
Interactive maps (raster/shapefile) with pan/zoom, layer toggling, and zoom-dependent detail.

**Timeline & Deliverables (2026)**
May: Initial wireframes and user journey mock-ups.
August: Functional Beta version with test API integration.
October: Final launch, bug fixing, and codebase handover for deployment.
**Budget:** 20.000 Euro (non-negotiable)

**Technical Stack & Requirements**

- **Framework:** Vue.js with the Vuetify UI component library
- **Language:** TypeScript (required for clean, type-safe, and scalable code)
- **Architecture:** Clean Architecture principles (ensures the codebase can be easily forked and customised by IIASA researchers)
- **Rendering:** Familiarity with Server-Side Rendering (SSR) via Nuxt.js (required for performance and SEO optimisation)

**Geospatial & Data Visualization**

- **Mapping Libraries:** OpenLayers or Leaflet (must support Raster and Shapefile overlays)
- **Data Formats:** Integration with GeoTIFF (COG), NetCDF, GeoJSON, CSV, and potentially Zarr
- **Charting:** Interactive data plots using D3.js, Chart.js, Highcharts, or similar
- **Base Maps:** Support for background map toggling and zoom-dependent data rendering

**Infrastructure & Integration**

- **API Consumption:** Interfacing with IIASA's cloud-hosted REST API that will provide the data to be displayed
- **Styling:** Modern CSS (e.g., Tailwind CSS) for responsive, mobile-friendly design
- **Version Control:** Git-based workflow (hosted on IIASA repositories)